

# chip\_lod: how many alternate reads make a CHIP call?

A plain-language guide to the model, the analyses and how to read them

**The question.** At one known position (for example a *DNMT3A* or *JAK2* hotspot), a sequencer reports  $n$  reads, and  $k$  of them show the variant base. Some of those reads can be sequencing errors. How many variant reads do we need before we can trust the call, and how small a variant can we detect at all?

**The answer in short.** With 1000 reads and an error rate of 0.001, a call needs at least **2 variant reads**. With that rule a variant at **0.44% VAF** or more is found in 95% of samples: that is the **limit of detection** (LoD). The LoD rises with the error rate (from 0.3% at  $e = 0.0001$  to 0.99% at  $e = 0.01$ ) and falls with depth. Detecting a 2% variant needs 148 reads, and a 0.5% variant 891 reads.

## Contents

|                                             |   |
|---------------------------------------------|---|
| 1. The question .....                       | 1 |
| 2. How the frequentist test works .....     | 1 |
| 3. Influence of the error rate .....        | 2 |
| 4. Influence of the read depth .....        | 3 |
| 5. Influence of the VAF .....               | 4 |
| 6. A function for the frequentist LoD ..... | 5 |
| 7. In practice: a panel at depth 1000 ..... | 5 |
| 8. Reproducibility .....                    | 7 |
| 9. Limitations .....                        | 8 |

## 1. The question

Clonal haematopoiesis of indeterminate potential (CHIP) is defined by a somatic mutation in blood at a variant allele fraction (VAF) of at least 2%. Blood cells are diploid and the variant sits on one of the two copies, so a VAF of 2% means about 4% of cells carry it.

Sequencing is not perfect. Even at a position where nobody carries the variant, a small fraction of reads show the variant base because of sequencing or PCR errors. With an error rate  $e = 0.001$  per base, and a third of errors landing on any particular wrong base, a site with 1000 reads is expected to show **0.333 error reads** on average. So when we see 1, 2 or 5 variant reads, is that a variant or noise?

This report answers it with classical (frequentist) statistics only: a significance test against the error background, and the limit of detection that follows from it. We deliberately do not use a Bayesian analysis, because that needs prior information (how common the variant is, and at what VAF) that we do not have.

All results come from the `chip_lod` package (Python, with an R port that produces byte-identical files). All settings are in `config/params.json`.

## 2. How the frequentist test works

The frequentist approach asks one question: **if there were no variant, how surprising would these reads be?** It needs only a model of the errors. Here is the procedure at depth 1000 with  $e = 0.001$ .

1. **Background.** Errors produce on average  $\lambda_0 = n \cdot e/3 = 0.333$  variant reads. The actual number varies from sample to sample and follows a Poisson distribution, the standard model for counts of rare, independent events.
2. **p-value.** The p-value of  $k$  reads is the probability that errors alone produce  $k$  or more variant reads. With  $k = 0$  the p-value is 1. With  $k = 1$  the p-value is 0.2835. With  $k = 2$  the p-value is 0.0446. With  $k = 5$  the p-value is  $2.6 \times 10^{-5}$ .

3. **Read-count limit of detection  $k^*$ .** We call the variant when the p-value is at most  $\alpha = 0.05$ , i.e. when errors would produce that many reads in fewer than 5% of variant-free samples. The smallest such count is  $k^* = 2$ . Because read counts are whole numbers, the actual false-positive rate is 0.0446, a little below  $\alpha$ .
4. **Power.** A real variant at VAF  $p$  produces variant reads at the rate  $r(p) = p(1 - e) + (1 - p)e/3$  per read (correct reads from the mutated copy plus misread normal reads). The **power** is the chance that such a variant reaches  $k^*$  reads and is called.
5. **Limit of detection.** The LoD is the smallest VAF that is called in at least 95% of samples. At depth 1000 this is **0.44%**.

**How to read this.**  $k^*$  is a property of the depth and the error rate, and is fixed before looking at the data. The LoD is a property of the assay: it says which variants the assay reliably detects, not whether this particular sample carries one. A variant below the LoD can still be called, just not reliably; a sample without a call can still carry a variant below the LoD.

The next sections vary the three ingredients one at a time (error rate, depth and VAF), focusing on VAFs up to 2%, the range where calling is difficult. Throughout,  $\alpha = 0.05$  and the required power is 95%.

### 3. Influence of the error rate

| error rate $e$ | background $\lambda_0$ | $k^*$ | actual false-positive rate | LoD VAF |
|----------------|------------------------|-------|----------------------------|---------|
| 0.0001         | 0.033                  | 1     | 0.0328                     | 0.3%    |
| 0.0005         | 0.167                  | 2     | 0.0124                     | 0.46%   |
| 0.001          | 0.333                  | 2     | 0.0446                     | 0.44%   |
| 0.002          | 0.667                  | 3     | 0.0302                     | 0.56%   |
| 0.005          | 1.667                  | 5     | 0.0275                     | 0.75%   |
| 0.01           | 3.333                  | 8     | 0.0208                     | 0.99%   |

Table 1: The effect of the error rate at depth 1000 (f1\_lod\_grid.csv).

A higher error rate means more background reads, so a call needs more variant reads ( $k^*$  rises from 1 to 8), and the LoD rises with it. Going from the cleanest assay ( $e = 0.0001$ , typical of UMI/duplex consensus reads) to the noisiest ( $e = 0.01$ ) raises the LoD about 3.4-fold.

The “actual false-positive rate” column shows the effect of whole-number thresholds. The test promises at most 5% false positives, but because  $k^*$  jumps in whole steps, the real rate is anywhere between almost 0 and 5%.

| depth | $e = 0.0001$ | $e = 0.0005$ | $e = 0.001$ | $e = 0.002$ | $e = 0.005$ | $e = 0.01$ |
|-------|--------------|--------------|-------------|-------------|-------------|------------|
| 100   | 2.99% (1)    | 2.98% (1)    | 2.97% (1)   | 4.69% (2)   | 4.61% (2)   | 4.47% (2)  |
| 200   | 1.49% (1)    | 1.48% (1)    | 2.34% (2)   | 2.31% (2)   | 2.22% (2)   | 2.85% (3)  |
| 500   | 0.6% (1)     | 0.93% (2)    | 0.92% (2)   | 0.88% (2)   | 1.39% (4)   | 1.52% (5)  |
| 1000  | 0.3% (1)     | 0.46% (2)    | 0.44% (2)   | 0.56% (3)   | 0.75% (5)   | 0.99% (8)  |
| 2000  | 0.23% (2)    | 0.22% (2)    | 0.28% (3)   | 0.32% (4)   | 0.49% (8)   | 0.58% (12) |
| 5000  | 0.09% (2)    | 0.14% (4)    | 0.15% (5)   | 0.2% (8)    | 0.25% (14)  | 0.35% (25) |
| 10000 | 0.04% (2)    | 0.07% (5)    | 0.1% (8)    | 0.12% (12)  | 0.17% (25)  | 0.22% (44) |

Table 2: LoD VAF for every combination of depth and error rate, with  $k^*$  in brackets (f1\_lod\_grid.csv).

## 4. Influence of the read depth

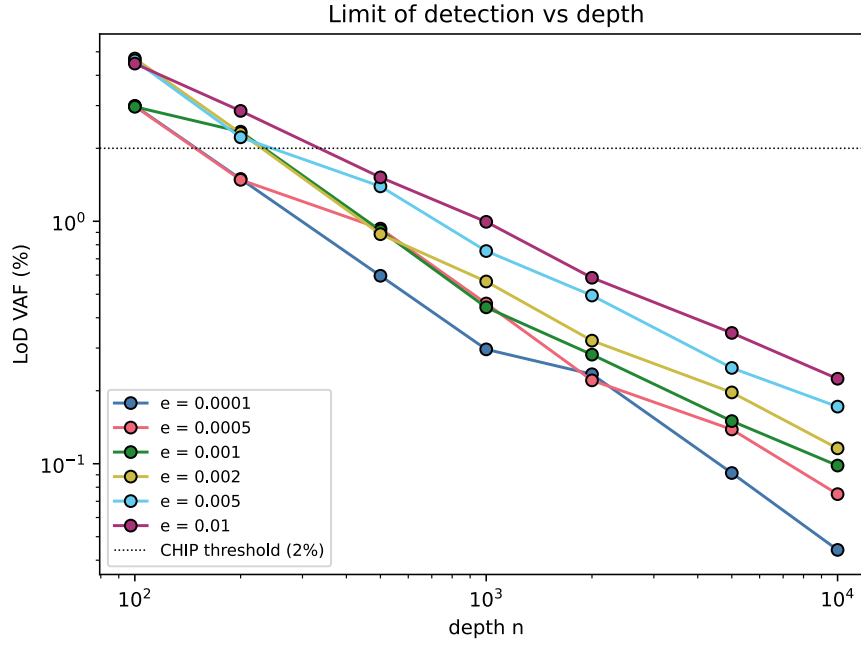


Figure 1: Limit of detection against depth for six error rates (log-log). The dotted line is the 2% CHIP threshold.

More reads make the variant reads stand out from the background, so the LoD falls with depth. While errors are negligible ( $k^*$  stays at 1 or 2), it falls almost in proportion to  $1/n$ : ten times the depth gives a ten times smaller LoD. Once the background amounts to several reads, the variant has to rise above the random scatter of those error reads, and the LoD falls more slowly, closer to  $1/\sqrt{n}$ . At  $e = 0.001$  the LoD goes from 2.97% at 100 reads to 0.1% at 10000 reads. The lines are not perfectly smooth: each time the depth grows enough for  $k^*$  to step up by one read, the LoD briefly gets worse before improving again. This is the “saw-tooth” effect of whole-number thresholds.

**How deep do I need to sequence?** The table below inverts the question. It gives the smallest depth at which a variant of a given VAF is detected with 95% power.

| error rate $e$ | VAF 0.5%            | VAF 1%            | VAF 2%            |
|----------------|---------------------|-------------------|-------------------|
| 0.0001         | 596 ( $k^* = 1$ )   | 299 ( $k^* = 1$ ) | 150 ( $k^* = 1$ ) |
| 0.0005         | 919 ( $k^* = 2$ )   | 295 ( $k^* = 1$ ) | 149 ( $k^* = 1$ ) |
| 0.001          | 891 ( $k^* = 2$ )   | 460 ( $k^* = 2$ ) | 148 ( $k^* = 1$ ) |
| 0.002          | 1114 ( $k^* = 3$ )  | 446 ( $k^* = 2$ ) | 231 ( $k^* = 2$ ) |
| 0.005          | 1786 ( $k^* = 7$ )  | 669 ( $k^* = 4$ ) | 293 ( $k^* = 3$ ) |
| 0.01           | 2501 ( $k^* = 14$ ) | 898 ( $k^* = 7$ ) | 337 ( $k^* = 4$ ) |

Table 3: Smallest depth with 95% power to detect each VAF (f2\_required\_depth.csv).

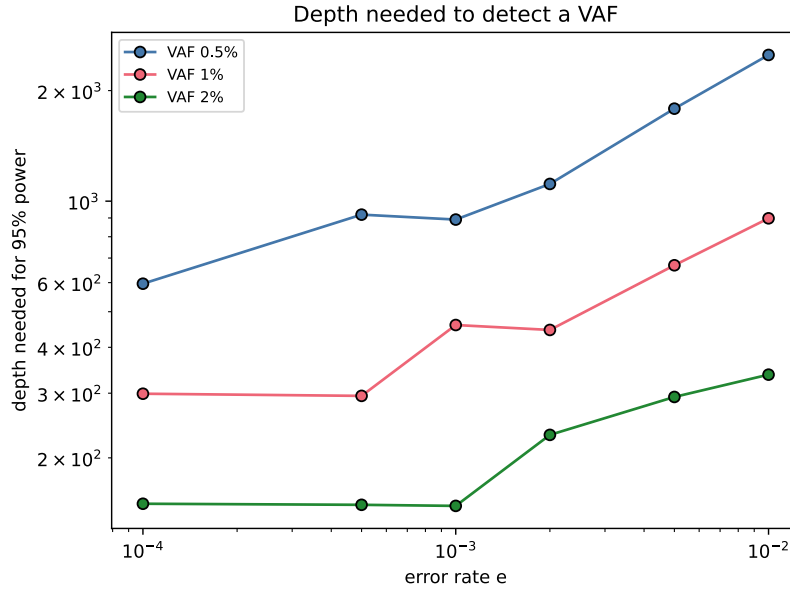


Figure 2: Depth needed for 95% power against error rate, for three target VAFs.

Small VAFs are expensive: halving the VAF roughly doubles (or more) the depth needed, and a noisy assay can need several times more reads than a clean one. One result looks paradoxical: for 0.5%, 1% and 2% VAF, a slightly *higher* error rate sometimes needs *fewer* reads. This happens when both error rates use the same  $k^*$ . The extra error reads then also count towards the variant's total and help it cross the threshold. The price is a higher actual false-positive rate (previous section), so the noisier assay is not really better.

## 5. Influence of the VAF

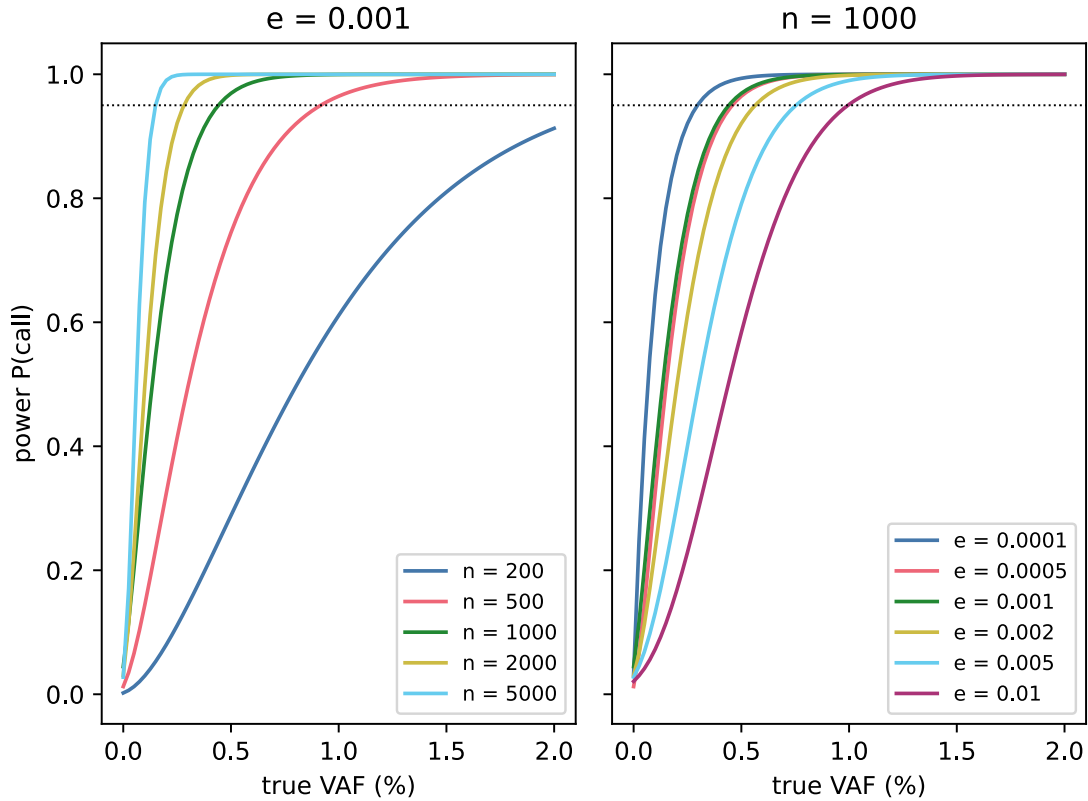


Figure 3: Power (probability of a call) against true VAF up to 2%. Left: depths at  $e = 0.001$ . Right: error rates at depth 1000. The dotted line is 95% power (f3\_power\_curve.csv).

| true VAF | n = 200 ( $k^* = 2$ ) | n = 500 ( $k^* = 2$ ) | n = 1000 ( $k^* = 2$ ) | n = 2000 ( $k^* = 3$ ) | n = 5000 ( $k^* = 5$ ) |
|----------|-----------------------|-----------------------|------------------------|------------------------|------------------------|
| 0.1%     | 3%                    | 14.4%                 | 38.4%                  | 49.8%                  | 79.4%                  |
| 0.25%    | 11.1%                 | 41.3%                 | 77.4%                  | 92.1%                  | 99.8%                  |
| 0.5%     | 28.8%                 | 74.5%                 | 96.9%                  | 99.8%                  | 100%                   |
| 1%       | 61.1%                 | 96.5%                 | 100%                   | 100%                   | 100%                   |
| 2%       | 91.3%                 | 100%                  | 100%                   | 100%                   | 100%                   |

Table 4: Probability of calling a variant of a given VAF at  $e = 0.001$  (selected rows of `f3_power_curve.csv`).

For a fixed assay, power rises steeply with VAF: below the LoD a variant is called only some of the time, and above it almost always. Between about 0.1% and 1% VAF, the region of interest for early clonal haematopoiesis, the answer depends strongly on depth. At  $e = 0.001$ , a variant at the 2% CHIP threshold is reliably detected from 148 reads onwards, 0.5% needs 891 reads, and 0.1% is first reached at 10000 reads in our depth grid.

## 6. A function for the frequentist LoD

`lod_frequentist(n, k, e, alpha, power)` applies the whole analysis to a list of sites. Given vectors of depths  $n$  and variant-read counts  $k$ , it returns for every site: the background  $\lambda_0$ , the p-value,  $k^*$ , whether the site is called, and the LoD at that depth.

```
from chip_lod.lod import lod_frequentist    source("r/load.R")
res = lod_frequentist(n=[500, 1000, 2000], lod_frequentist(n = c(500, 1000, 2000),
k=[3, 1, 3], e=0.001)                      k = c(3, 1, 3), e = 0.001)
res["called"], res["lod_vaf"]              # data.frame: n, k, lambda_bg, pvalue,
#                                         k_star, called, lod_vaf
```

| $n$  | $k$ | $\lambda_0$ | p-value              | $k^*$ | called | LoD VAF |
|------|-----|-------------|----------------------|-------|--------|---------|
| 500  | 3   | 0.167       | $6.8 \times 10^{-4}$ | 2     | yes    | 0.92%   |
| 1000 | 0   | 0.333       | 1                    | 2     | no     | 0.44%   |
| 1000 | 1   | 0.333       | 0.2835               | 2     | no     | 0.44%   |
| 1000 | 2   | 0.333       | 0.0446               | 2     | yes    | 0.44%   |
| 1000 | 5   | 0.333       | $2.6 \times 10^{-5}$ | 2     | yes    | 0.44%   |
| 2000 | 3   | 0.667       | 0.0302               | 3     | yes    | 0.28%   |

Table 5: Output of `lod_frequentist` for example sites at  $e = 0.001$  (`f5_lod_function_example.csv`).

**Methods paragraph** (for a paper; also in `spec/methods_lod_frequentist.txt`):

Limit of detection. Sequencing errors can produce reads that look like a variant, so we first estimated how many such reads to expect at a site without a variant. Only about one in three base-calling errors produces the specific variant base, so a site sequenced to depth  $n$  with a per-base error rate  $e$  is expected to carry  $n \cdot e / 3$  error reads, and we modelled their number with a Poisson distribution. From this we derived, for each depth, the read-count limit of detection  $k^*$ : the smallest number of variant reads that errors alone produce in fewer than 5% of variant-free samples. A variant was called when it was supported by at least  $k^*$  reads. A real heterozygous variant at variant allele fraction (VAF)  $p$  contributes variant reads at a rate of  $p(1 - e) + (1 - p) \cdot e / 3$ , close to  $p$ , which tells us how often it reaches  $k^*$  reads. The VAF limit of detection is the lowest VAF that reaches  $k^*$  reads in at least 95% of samples. Because read counts are whole numbers, the false-positive rate is at most 5% and the limit of detection does not decrease smoothly with depth. All values were computed with the function `lod_frequentist` (Python and R).

## 7. In practice: a panel at depth 1000

A typical targeted panel gives about 1000 reads at a hotspot, and the depth varies from sample to sample. At  $e = 0.001$  the background is 0.333 error reads.

| variant reads $k$ | p-value              | called at $\alpha = 0.05$ ? | called at $\alpha = 0.001$ ? |
|-------------------|----------------------|-----------------------------|------------------------------|
| 0                 | 1                    | no                          | no                           |
| 1                 | 0.2835               | no                          | no                           |
| 2                 | 0.0446               | yes                         | no                           |
| 3                 | 0.0048               | yes                         | no                           |
| 4                 | $3.9 \times 10^{-4}$ | yes                         | yes                          |
| 5                 | $2.6 \times 10^{-5}$ | yes                         | yes                          |
| 6                 | $1.4 \times 10^{-6}$ | yes                         | yes                          |
| 7                 | $6.8 \times 10^{-8}$ | yes                         | yes                          |
| 8                 | $2.8 \times 10^{-9}$ | yes                         | yes                          |

Table 6: p-values for 0–8 variant reads at depth 1000 (a3\_read\_table.csv, which runs to  $k = 30$ ).

| depth | $\alpha = 0.05$ |                |         | $\alpha = 0.001$ |                      |         |
|-------|-----------------|----------------|---------|------------------|----------------------|---------|
|       | $k^*$           | actual FP rate | LoD VAF | $k^*$            | actual FP rate       | LoD VAF |
| 800   | 2               | 0.0298         | 0.56%   | 4                | $1.7 \times 10^{-4}$ | 0.94%   |
| 1000  | 2               | 0.0446         | 0.44%   | 4                | $3.9 \times 10^{-4}$ | 0.74%   |
| 1200  | 3               | 0.0079         | 0.49%   | 4                | $7.8 \times 10^{-4}$ | 0.61%   |

Table 7: Read-count limit  $k^*$ , actual false-positive rate and LoD around depth 1000 for a lenient and a strict  $\alpha$  (a3\_thresholds.csv).

A stricter significance level protects better against false calls but costs sensitivity: at depth 1000,  $\alpha = 0.001$  needs 4 reads instead of 2 and raises the LoD from 0.44% to 0.74%. The LoD does not always fall as depth rises. When  $k^*$  steps up by one read, the LoD can jump up slightly (here from 0.44% at depth 1000 to 0.49% at depth 1200,  $\alpha = 0.05$ ).

**Simulated panel.** We simulated sites whose depth varies uniformly between 800 and 1200, and called each one with its own  $k^*$  ( $\alpha = 0.05$ ).

| true VAF | sites | mean depth | call rate | analytic power at 1000 |
|----------|-------|------------|-----------|------------------------|
| 0%       | 500   | 994        | 3.2%      | 4.5%                   |
| 0.1%     | 500   | 1004       | 31.4%     | 38.4%                  |
| 0.25%    | 500   | 1005       | 70.6%     | 77.4%                  |
| 0.5%     | 500   | 999        | 95.2%     | 96.9%                  |
| 1%       | 500   | 1006       | 99.8%     | 100%                   |
| 2%       | 500   | 995        | 100%      | 100%                   |
| 5%       | 500   | 1005       | 100%      | 100%                   |

Table 8: Call rates in simulated data with variable depth (a3\_sim\_sensitivity.csv). The VAF 0 row is the false-positive rate.

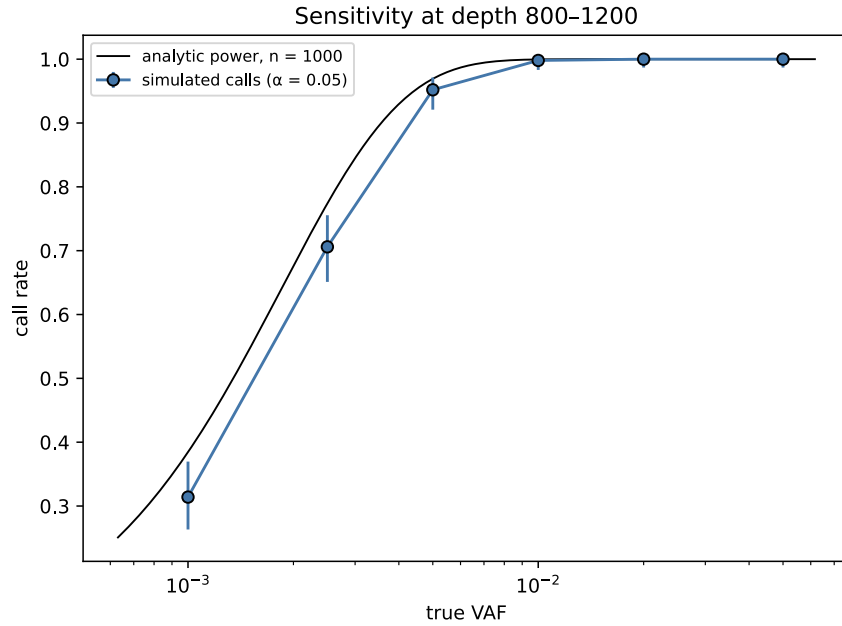


Figure 4: Call rate against true VAF for sites with variable depth (99% Wilson intervals), with the analytic power at depth 1000.

With no variant, 3.2% of sites were called, below  $\alpha = 0.05$ . The analytic power assumes exactly 1000 reads. Deeper simulated sites need more reads ( $k^* = 3$  at depth 1200), which is why the simulated call rates sit slightly below it: when depth varies between samples, the LoD of the panel is set by its deeper sites as well as its average.

## 8. Reproducibility

The package has two independent implementations, Python and R, written to produce **byte-identical** result files. Both use the same hand-written random number generator, explicit left-to-right sums and the same number formatting. `scripts/compare_outputs.sh` compares the SHA-256 hash of every file.

**Python and R agree:** 11 of 11 result files have identical SHA-256 hashes.

| file                        | SHA-256 (first 16 hex digits) | match |
|-----------------------------|-------------------------------|-------|
| a3_read_table.csv           | afadec0a43527ef4              | yes   |
| a3_sim_sensitivity.csv      | 0b64d0021e446bd7              | yes   |
| a3_thresholds.csv           | 36b6c91829ac1bb4              | yes   |
| f1_lod_grid.csv             | 00b4d11df4f61be7              | yes   |
| f2_required_depth.csv       | b4e9249abbf100fc              | yes   |
| f3_power_curve.csv          | b07ff7605f6e126d              | yes   |
| f4_sim_power.csv            | e9588c9c469b8cb4              | yes   |
| f5_lod_function_example.csv | 26b186538aa1f161              | yes   |
| sim_a3.csv                  | 99f2ba0917c05a33              | yes   |
| sim_f.csv                   | 1d9bd914c6faab92              | yes   |
| summary.json                | 4784aa6b52e2dc9c              | yes   |

Every number in this report is read from `results/python/` when the document is compiled, so rerunning `make` all updates everything. The model is specified in `spec/MODEL.md`, and every judgement call is listed in `spec/DECISIONS.md`.

## 9. Limitations

- **Poisson approximation.** Read counts are really binomial, and the Poisson is slightly too wide. The simulations show the difference is negligible at these depths and VAFs.
- **No overdispersion.** Real error counts vary more from sample to sample than a Poisson allows (batch, library and position effects). A beta-binomial or negative-binomial background would give larger, more honest  $k^*$  values.
- **The error rate is site- and context-specific.** A single  $e = 0.001$  is a placeholder. Error rates depend on the base change, the sequence context and the assay; the error-rate section shows how strongly this matters. In practice, estimate the background at each hotspot from a **panel of normals**.
- **PCR duplicates and UMIs.** The model treats reads as independent. Duplicates from the same molecule are not independent and make a variant look better supported than it is. Count unique molecules (UMI families), not raw reads; UMI consensus also lowers  $e$  considerably.
- **Diploid, heterozygous assumption.**  $\text{VAF} = \text{cell fraction} / 2$  assumes a heterozygous variant on a normal copy number. Loss of heterozygosity, copy-number changes or a homozygous variant break this link.